

CMPS 2020: Latino presidential vote

Three nested random forests and item-level SHAP

Question

Which survey responses distinguish reported Trump voters from other presidential voters among Latinos in the 2020 CMPS? Three nested models describe the full predictive structure, then the structure remaining after withholding information close to the electoral choice and direct partisan attachment. This post-election analysis describes predictive associations, not causes of vote choice.

Sample and outcome

The sample comprises **2,082 Latino registered respondents in the primary release** who reported voting and named a valid presidential choice. It includes **568 Trump voters: 27.28%** unweighted and **28.33%** weighted. The other reported choices are Biden, Jorgensen, Hawkins, or someone else. Respondents who answered a nonvoter candidate-support version are excluded.

$$Y_i = \begin{cases} 1 & \text{Trump vote} \\ 0 & \text{other reported presidential vote} \end{cases}$$

The within-racial-group **WEIGHT** targets the 2019 ACS adult population using age, gender, education, nativity, and ancestry margins. The primary-sample methodology recommends this weight for multivariate analysis. It is applied to the voter subset without trimming or class balancing; it is not a separate calibration to Latino voters. The primary-release restriction excludes 138 otherwise eligible quota-group voters whose weight scope is not established by that documentation. Exact definitions and release discrepancies appear in the audit.

Three nested models

Model	Information available	Purpose
1. Full	All otherwise eligible survey responses	Describe the full predictive structure
2. Outcome-proximate withheld	Model 1 minus candidate and election-proximate responses	Describe differentiation beyond information close to vote choice
3. Direct partisanship withheld	Model 2 minus direct partisan information; ideology retained	Describe differentiation beyond explicit partisan attachment

Model 2 withholds candidate evaluations, congressional vote, presidential-election fraud and January 6 reactions, candidate-framed anxiety and policy evaluations, and hypothetical candidate-response experiments. Model 3 additionally withholds party identification, strength, leaning, history, party associations and affinity, party-selected identity responses, partisan redistricting tradeoffs, and spouse's partisanship. Generic political activity and policy attitudes remain eligible. No exclusion depends on an item's fitted importance.

Why random forest + SHAP

The random forest estimates $\hat{p}_i = f(X_i)$ and allows nonlinear relationships and interactions without specifying their forms beforehand. TreeSHAP decomposes its predictions relative to a training reference:

$$\hat{p}_i \approx \phi_0 + \sum_j \phi_{ij}.$$

Weighted mean absolute SHAP, $I_j = E_w(|\phi_{ij}|)$, measures an item’s overall predictive contribution. Contributions from encoded columns are summed back to the original item **before** taking absolute values. Signed SHAP describes responses associated with predictions above or below the model reference. Neither is a causal effect. Correlated items can share or substitute for one another’s predictive contributions.

The complete outer training sample supplies the unweighted TreeSHAP reference for tree-path frequencies. Survey weights enter forest resampling, evaluation, global importance, and response-level summaries. No individual predictions or individual SHAP arrays are exported.

Validation and stability

Three outcome-stratified 75/25 outer splits use seeds **202001**, **202002**, **202003**. Each outer training portion has an 80/20 internal training/validation split. Nine candidates cross $\lfloor \sqrt{p} \rfloor$, $\lfloor 0.20p \rfloor$, and $\lfloor 0.40p \rfloor$ with node sizes 5, 20, and 50; p is the encoded feature count after the relevant training-only preprocessing. Weighted validation Brier loss selects a setting using 150 trees; the final forest uses 300 trees. The grid is frozen even when 40% is selected.

All screening and preprocessing use only the relevant training portion: at least 30 observed responses, at most 90% system missingness, variation checks, category pooling below 10 training cases, numeric median imputation, and removal of constant encoded columns. Numeric missingness indicators belong to their source item. Categorical system missingness has an explicit “Not observed / routed” category. Original labelled responses—including don’t know, refused, none, and unfamiliar—remain distinct unless pooled by the training rule. There is no blanket deletion of numeric codes and no reverse scoring.

Randomized wording is joined to the relevant answer. Rating categories retain original orientation; numeric opinion scales and ranks are not automatically continuous. Continuous sliders, point allocations, birth year, and verified calendar-year quantities retain their magnitudes. Signed summaries use original responses, with at least 10 holdout observations per cell. Continuous measures with more than 12 distinct values use training-derived quartile bins.

The holdouts never select predictors, transformations, or tuning settings. They overlap across fits: these are stability checks, not independent replications, and ranges are **not confidence intervals**. Calibration is diagnostic; fitted probabilities are not presented as individual political scores.

2020 results

Performance is reported as mean [minimum, maximum] across three weighted outer holdouts. Smaller Brier loss indicates smaller prediction error. The reference Brier uses the outer-training weighted Trump prevalence.

Model	AUC	Brier	Reference
Model 1	0.989 [0.987, 0.993]	0.037 [0.033, 0.040]	0.204 [0.200, 0.212]
Model 2	0.967 [0.965, 0.968]	0.070 [0.064, 0.077]	0.204 [0.200, 0.212]
Model 3	0.936 [0.922, 0.949]	0.092 [0.083, 0.102]	0.204 [0.200, 0.212]

Model 1 — Full

Congressional vote, Trump immigration-policy approval, and Trump favorability rank first, second, and third in every split. Their mean absolute contributions are 15.44, 6.56, and 4.23 percentage points. Trump responsibility for COVID-19 and January 6, other candidate evaluations, and hypothetical presidential choices follow. This strong predictive separation relies heavily on information close to the reported electoral choice.

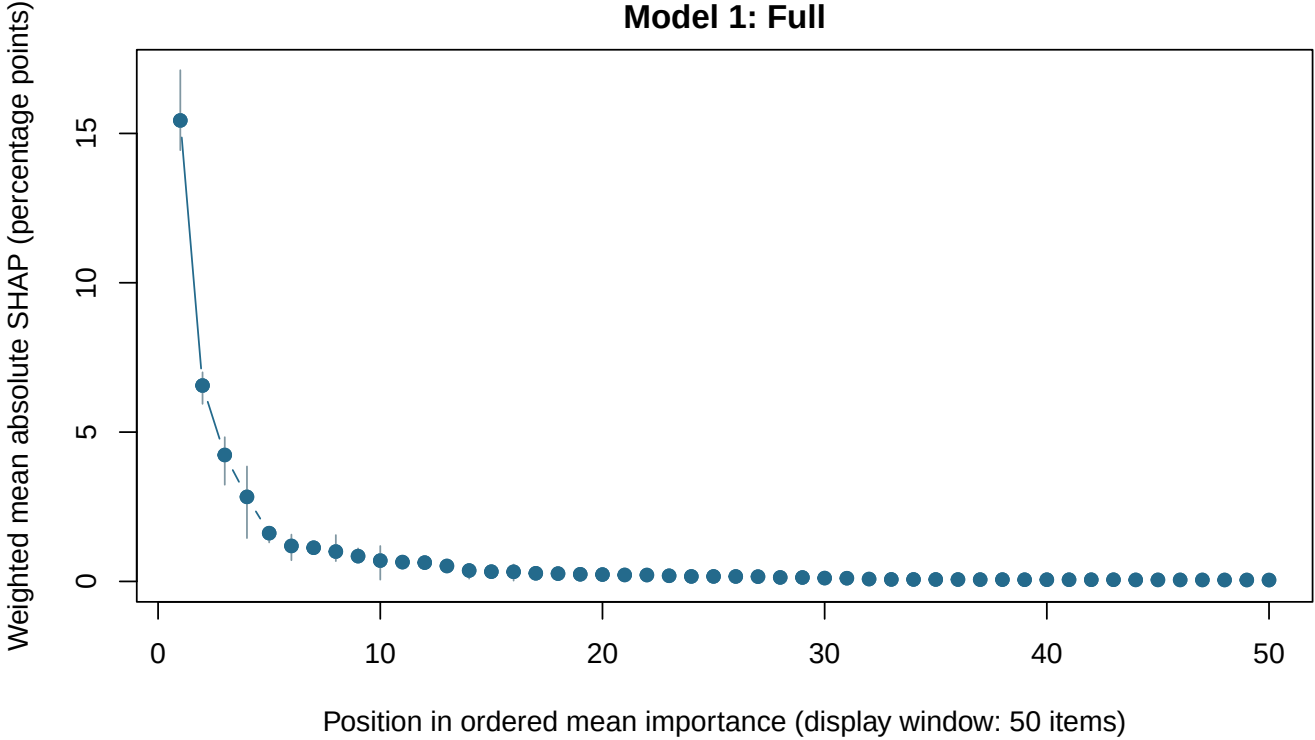


Figure 1: Model 1: top-50 importance profile. Vertical lines show the across-fit range.

Position	Item	SHAP (pp)	Rank	Range	Fits	Gap (pp)
1	Q15: Congressional vote	15.435	1.0	1–1	3	8.876
2	Q614: Trump immigration-policy approval	6.559	2.0	2–2	3	2.326
3	Q2R1: Trump favorability	4.233	3.0	3–3	3	1.403
4	Q130: Trump responsibility for COVID-19	2.830	4.3	4–5	3	1.214
5	Q54: Trump responsibility for January 6	1.616	5.3	5–6	3	0.428
6	Q2R3: Biden favorability	1.188	7.7	6–11	3	0.060
7	Q195A2: Ranked-choice vignette (short, 10 candidates)	1.128	7.0	7–7	3	0.128
8	Q195A1: Ranked-choice vignette (short, 4 candidates)	1.000	7.7	4–10	3	0.153
9	Q195B2: Ranked-choice vignette (long, 10 candidates)	0.847	9.3	8–11	3	0.149
10	Q2R4: Harris favorability	0.698	17.0	8–35	3	0.051
11	Q195B1: Ranked-choice vignette (long, 4 candidates)	0.646	11.3	9–15	3	0.016
12	Q197A1: Single-choice vignette (short, 4 candidates)	0.630	12.0	10–14	3	0.114

Position	Item	SHAP (pp)	Rank	Range	Fits	Gap (pp)
13	Q197B1: Single-choice vignette (long, 4 candidates)	0.516	12.7	12–14	3	0.151
14	Q21: Party identification	0.365	15.7	13–21	3	0.038
15	Q52: Fraud claim after Biden certification	0.327	16.7	13–19	3	0.005
16	Q24: Republican party history	0.322	35.7	11–80	3	0.050
17	Q197B2: Single-choice vignette (long, 10 candidates)	0.272	19.3	15–26	3	0.011
18	Q197A2: Single-choice vignette (short, 10 candidates)	0.262	20.0	14–25	3	0.022
19	Q25: Democratic party history	0.239	48.7	16–110	3	0.010
20	Q2R7: Hillary Clinton favorability	0.230	23.0	15–33	3	0.013
21	Q2R9: Michelle Obama favorability	0.217	20.0	19–22	3	0.003
22	Q22: Partisan strength	0.214	21.3	18–24	3	0.025
23	Q51: Fraud claim after election	0.189	23.0	18–28	3	0.021
24	Q2R8: Melania Trump favorability	0.167	24.7	17–29	3	0.000
25	Q308R7: BLM warmth	0.167	24.3	16–30	3	0.002
26	Q1: Issue priorities (eligible components)	0.165	37.3	20–65	3	0.005
27	Q55: Republican challenge to certification	0.160	32.7	17–51	3	0.026
28	Q2R5: Barack Obama favorability	0.134	48.0	24–94	3	0.002
29	Q2R10: Jill Biden favorability	0.132	26.3	20–36	3	0.020
30	Q504R11: Anxiety from Trump	0.112	28.0	25–31	3	0.004

Model 2 — Outcome-proximate measures withheld

Party identification and Republican and Democratic party histories occupy the first three positions in mean importance, although their individual ranks vary. Border-security spending and issue priorities follow. Associations of groups with the two parties become especially prominent in one split, so a fixed order among partisan measures is not stable. BLM support, BLM warmth, and the most-disliked-group ranking also remain prominent. Routed party histories repeat information about attachment and should not be interpreted as independent partisan mechanisms.

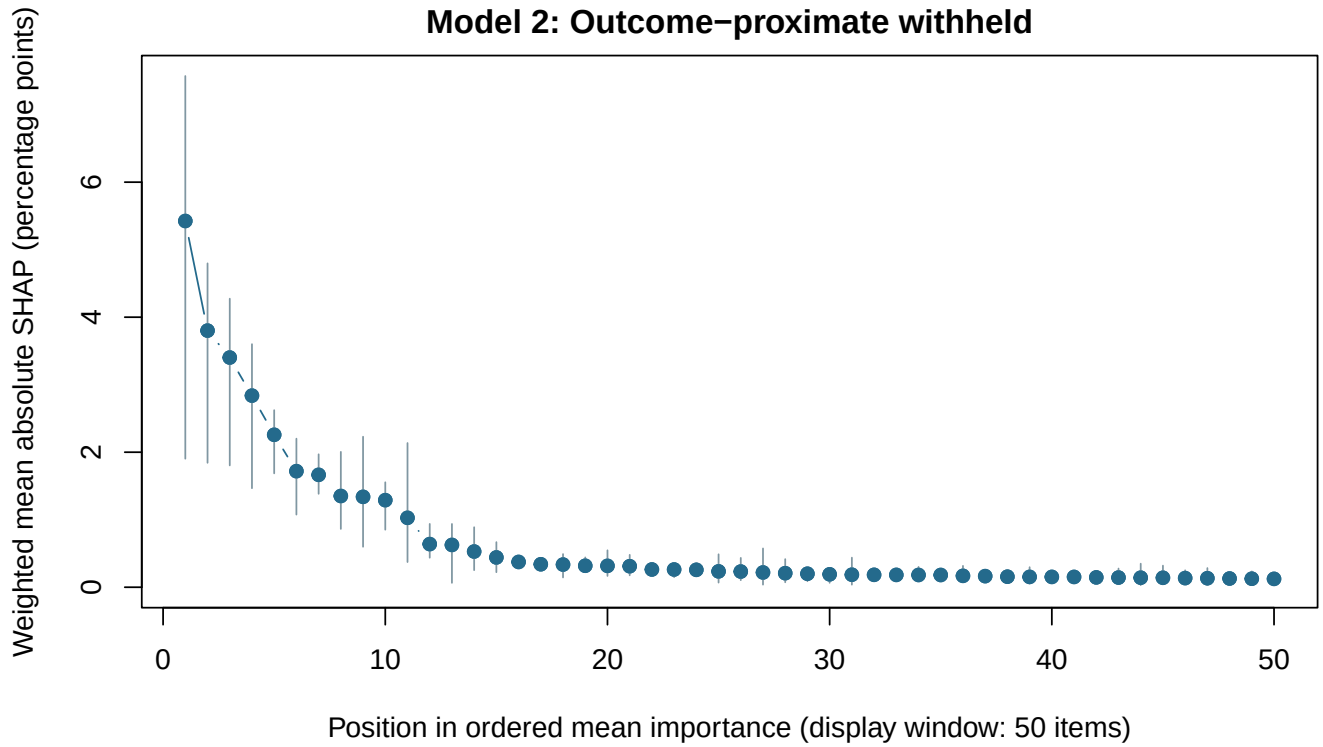


Figure 2: Model 2: top-50 profile. The display window is not a substantive cutoff.

Position	Item	SHAP (pp)	Rank	Range	Fits	Gap (pp)
1	Q21: Party identification	5.425	1.7	1–3	3	1.624
2	Q24: Republican party history	3.801	2.7	2–4	3	0.399
3	Q25: Democratic party history	3.402	3.7	3–5	3	0.564
4	Q543: Border-security spending and wall	2.838	5.3	4–8	3	0.580
5	Q1: Issue priorities (eligible components)	2.258	5.3	5–6	3	0.539
6	Q22: Partisan strength	1.719	7.7	7–9	3	0.054
7	Q606: Most-disliked groups: ranking	1.665	7.3	6–9	3	0.314
8	Q28: Groups associated with Republicans	1.351	8.3	2–13	3	0.012
9	Q308R7: BLM warmth	1.339	9.0	6–12	3	0.049
10	Q358: BLM movement support	1.290	8.7	8–10	3	0.261
11	Q27: Groups associated with Democrats	1.028	9.3	1–15	3	0.389
12	Q607_Q610R2: Threat to American way of life (least-liked group)	0.639	14.7	10–21	3	0.012
13	Q23: Party leaning	0.627	44.7	11–112	3	0.098
14	Q504R6: Anxiety from white supremacy	0.529	21.3	12–38	3	0.087
15	Q485: Racial bias in police/immigration treatment	0.441	16.0	11–22	3	0.067
16	Q607_Q610R4: Ban candidacy (least-liked group)	0.374	16.7	16–18	3	0.034
17	Q607_Q610R1: Threat to family safety (least-liked group)	0.341	20.0	14–27	3	0.005
18	Q308R6: MeToo warmth	0.336	20.7	15–30	3	0.018
19	Q308R5: Climate activists warmth	0.318	19.7	19–20	3	0.001
20	Q43: Ideology	0.317	20.3	14–26	3	0.005
21	Q184R1: Border priority despite migrant deaths	0.311	20.0	17–25	3	0.049

Position	Item	SHAP (pp)	Rank	Range	Fits	Gap (pp)
22	Q607_Q610R3: Ban public rallies (least-liked group)	0.263	24.0	19–32	3	0.000
23	Q308R2: Gun owners warmth	0.263	22.7	17–27	3	0.005
24	Q225R11: Republicans: support/threat to own vision	0.258	23.7	22–25	3	0.023
25	Q743: Spouse’s party identification	0.235	37.7	16–68	3	0.003
26	Q33R3: CNN news exposure	0.232	31.3	22–44	3	0.013
27	Q563_Q572R5: Whites: political representation	0.219	75.0	13–114	3	0.011
28	Q706_Q708R1: Shift police funds to community	0.208	37.7	23–60	3	0.008
29	Q208R5: Government reducing income differences	0.200	38.3	18–49	3	0.008
30	Q563_Q572R3: Immigrants: political representation	0.191	41.0	23–67	3	0.006

Model 3 — Direct partisanship withheld

Border-security spending and BLM support rank first and second in every split, with mean absolute contributions of 7.41 and 5.00 percentage points. The most-disliked-group ranking, issue priorities, and BLM warmth occupy positions 3–5 in each fit. Ideology follows at ranks 6–7. Racial bias in policing and immigration enforcement, gun-owner warmth, and the border-priority tradeoff are also recurrent. These responses describe differentiation beyond explicit party attachment; they are not necessarily independent of partisanship.

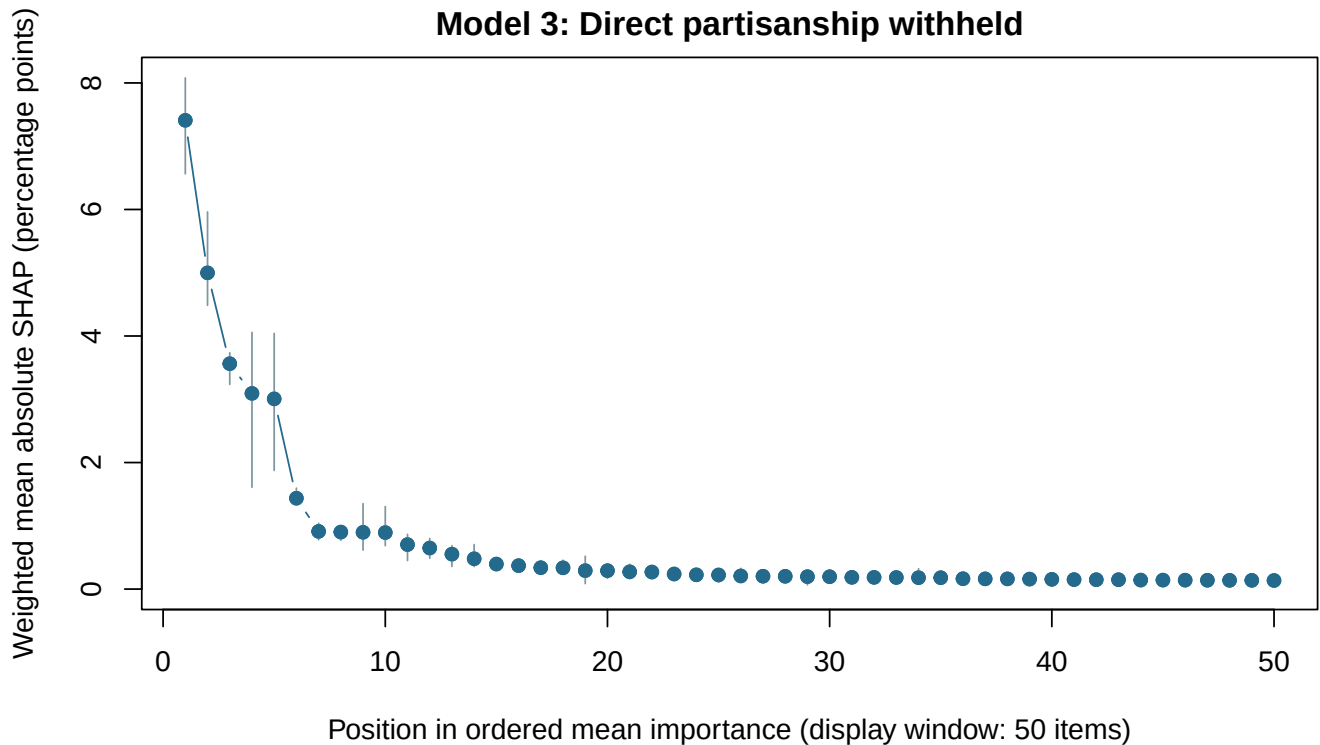


Figure 3: Model 3: top-50 profile. All retained items are ranked in the aggregate outputs.

Position	Item	SHAP (pp)	Rank	Range	Fits	Gap (pp)
1	Q543: Border-security spending and wall	7.409	1.0	1–1	3	2.411
2	Q358: BLM movement support	4.998	2.0	2–2	3	1.436
3	Q606: Most-disliked groups: ranking	3.562	3.7	3–4	3	0.469
4	Q1: Issue priorities (eligible components)	3.093	4.0	3–5	3	0.086
5	Q308R7: BLM warmth	3.007	4.3	3–5	3	1.570
6	Q43: Ideology	1.436	6.3	6–7	3	0.525
7	Q485: Racial bias in police/immigration treatment	0.911	8.7	7–10	3	0.010
8	Q308R2: Gun owners warmth	0.901	8.3	7–10	3	0.004
9	Q607_Q610R2: Threat to American way of life (least-liked group)	0.897	9.3	6–12	3	0.004
10	Q184R1: Border priority despite migrant deaths	0.893	10.3	8–12	3	0.190
11	Q706_Q708R1: Shift police funds to community	0.703	10.7	8–15	3	0.054
12	Q563_Q572R10: LGBTQ: political representation	0.649	11.3	9–13	3	0.097
13	Q563_Q572R9: Muslims: political representation	0.552	14.0	11–18	3	0.073
14	Q308R5: Climate activists warmth	0.479	15.0	11–18	3	0.084
15	Q607_Q610R4: Ban candidacy (least-liked group)	0.395	15.7	13–17	3	0.025
16	Q308R6: MeToo warmth	0.370	16.7	15–19	3	0.033
17	Q607_Q610R3: Ban public rallies (least-liked group)	0.338	19.3	16–24	3	0.002
18	Q607_Q610R1: Threat to family safety (least-liked group)	0.336	19.7	14–24	3	0.044
19	Q563_Q572R3: Immigrants: political representation	0.292	35.0	14–70	3	0.001
20	Q504R6: Anxiety from white supremacy	0.291	24.3	17–35	3	0.018
21	Q385: Defunding police: too far/not far enough	0.273	24.7	14–35	3	0.004
22	Q563_Q572R5: Whites: political representation	0.269	23.7	15–33	3	0.031
23	Q131R3: Ending parent-child border separation	0.238	29.7	20–46	3	0.013
24	Q217R1: White racial advantages	0.225	27.0	24–31	3	0.002
25	Q144: COVID-19 vaccine intentions	0.223	33.0	22–55	3	0.016
26	Q60: Theological self-identification	0.207	32.7	19–41	3	0.004
27	Q709_712R1: White privilege as major problem	0.203	33.3	20–43	3	0.001
28	Q619_Q626R5: Discrimination against immigrants	0.202	30.3	28–32	3	0.005
29	Q359R2: Own group benefits from BLM	0.197	43.0	20–86	3	0.001
30	Q504R3: Anxiety from climate change	0.195	31.7	22–39	3	0.010

How the prominent items behave

Importance and stability

The three tables show the first 30 positions. All-item outputs include each fit’s importance and rank, across-fit means and ranges, number of fits retained, and the adjacent mean-importance gap $\Delta_k = I_k - I_{k+1}$. Means weight fits equally

after weighting within each holdout; item means are conditional on retention. The complete eligible item-by-fit grid records nonretention as missing, not zero. Gaps are descriptive, not significance tests. The top-50 profiles show the shape of the distribution without selecting a final substantive threshold.

The sharpest stable ordering is at the top of Models 1 and 3. Below these leading positions, rank ranges often widen. In Model 2, party identification ranges from rank 1 to 3, but Democratic-group associations range from 1 to 15 and party leaning from 11 to 112. This variation cautions against treating small differences in average rank as fixed substantive distinctions. All displayed items are retained in all three fits; the complete output also records less consistently retained items.

Signed responses and nonlinear patterns

Positive signed SHAP moves a fitted prediction toward Trump vote relative to the reference; negative values move it away. Each response mean is weighted within fit, then averaged across fits. Tables require at least 10 holdout observations in each of all three fits. “N per fit” gives the observed range; the saved `direction.csv` records the count separately for each fit. For grouped questions, a component response conditions the whole item’s contribution, not an isolated component effect. Response labels below preserve the supplied categories; verified expansions of truncated labels are documented separately.

Model 1: Full — Congressional vote (Q15)

Response	SHAP (pp)	Range (pp)	N per fit
Republican candidate	26.31	23.28 to 30.15	150–164
Democratic candidate	-10.70	-11.91 to -10.05	349–363

Model 1: Full — Trump immigration-policy approval (Q614)

Response	SHAP (pp)	Range (pp)	N per fit
I oppose Trump’s immigration policies	-4.86	-5.37 to -4.25	364–369
I support Trump’s immigration policies	10.38	9.37 to 11.01	152–157

Model 1: Full — Trump favorability (Q2R1)

Response	SHAP (pp)	Range (pp)	N per fit
Very favorable	6.56	5.59 to 7.46	74–86
Somewhat favorable	4.66	3.12 to 6.61	62–66
Somewhat unfavorable	2.98	1.95 to 4.91	34–40
Very unfavorable	-3.75	-4.77 to -2.87	324–331
No opinion	1.78	0.01 to 3.64	10–14

Model 2: Outcome-proximate withheld — Party identification (Q21)

Response	SHAP (pp)	Range (pp)	N per fit
Republican	11.70	3.89 to 15.76	108–111
Democrat	-4.96	-6.84 to -1.77	248–280
Independent	0.57	-0.41 to 1.73	123–145
Other party	0.09	-0.58 to 0.46	10–20

Model 3: Direct partisanship withheld — Border-security spending and wall (Q543)

Response	SHAP (pp)	Range (pp)	N per fit
Support government increasing spending by \$25 billion on border security, including building a wall between the U.S. and Mexico	10.49	9.56 to 11.3	182–204
Oppose government increasing spending by \$25 billion on border security, including building a wall between the U.S. and Mexico	-5.46	-5.93 to -4.62	317–339

Model 3: Direct partisanship withheld — BLM movement support (Q358)

Response	SHAP (pp)	Range (pp)	N per fit
Strongly support	-3.31	-4.1 to -2.81	143–164
Somewhat support	-3.58	-4.29 to -3.02	106–127
Neither support nor oppose	-3.35	-4.15 to -2.8	100–115
Somewhat oppose	0.58	-1.01 to 1.85	27–49
Strongly oppose	11.81	10.3 to 13.86	94–106

Model 3: Direct partisanship withheld — Ideology (Q43)

Response	SHAP (pp)	Range (pp)	N per fit
Very Liberal	-0.75	-0.79 to -0.7	66–72
Somewhat Liberal	-1.10	-1.19 to -1.02	114–120
Moderate	-0.71	-0.82 to -0.63	179–185
Somewhat Conservative	0.78	0.56 to 0.93	63–77
Very Conservative	5.56	5.14 to 6.14	54–68
None of these	-0.54	-0.56 to -0.53	14–20

Model 3: Direct partisanship withheld — Dislike ranking: White supremacists (Q606R1)

Response	SHAP (pp)	Range (pp)	N per fit
1st	-2.09	-2.17 to -1.97	314–332
2nd	2.48	2.08 to 2.71	62–73
3rd	4.40	3.64 to 5.15	32–49
Not observed / routed	5.94	5.55 to 6.4	85–95

Model 3: Direct partisanship withheld — Dislike ranking: BLM activists (Q606R3)

Response	SHAP (pp)	Range (pp)	N per fit
1st	6.85	5.46 to 7.77	46–62
2nd	5.01	4.33 to 5.4	47–55
3rd	4.62	4.2 to 4.88	36–60
Not observed / routed	-1.94	-2.28 to -1.51	366–369

The signed patterns refine the interpretation of importance. In Model 1, Trump favorability is roughly ordered, but even “somewhat unfavorable” has a positive contribution relative to the reference; “very unfavorable” is the consistently

negative category. The item should not be reduced to a favorable versus unfavorable contrast. Trump immigration-policy opposition and support have consistently opposite signs, as do Democratic and Republican congressional votes.

In Model 2, Republican identification contributes positively and Democratic identification negatively in every fit. Independent and other-party means change sign across fits and are much closer to zero.

In Model 3, support for the specified border-security spending and wall is positive in every fit; opposition is negative. BLM support shows an **extreme-category pattern**, not a smooth support-to-opposition gradient: strong opposition contributes +11.81 percentage points on average [10.30, 13.86], whereas somewhat opposing averages +0.58 [−1.01, 1.85], with an unstable sign. Strong support, somewhat support, and neither support nor oppose all have negative means. The same strong-opposition distinction appears in Model 2 (+3.18 [2.20, 3.82]); somewhat opposing is near zero and unstable there too.

Very conservative ideology contributes much more positively than somewhat conservative ideology; liberal and moderate responses are negative. This is a broad directional pattern with a pronounced conservative endpoint, rather than equal steps across an ordinal scale.

The most-disliked-group ranking is also differentiated: placing White supremacists first is associated with a negative whole-item contribution, while placing them second or third is positive. Ranking BLM activists among the three most disliked groups is positive in every displayed rank. These are conditional summaries of the complete ranking, not separate effects of moving a group.

Selecting border security as an issue priority (Q1R19) accompanies a +9.71 percentage-point mean for the **whole priority item** [4.70, 12.92; 80–83 cases per fit], versus −1.78 [−2.25, −0.88; 438–441 cases] when not selected. This is not an isolated contribution of that checkbox. On anxiety from White supremacy (Q504R6, 1 = least, 10 = most), the lowest category contributes +0.94 [0.56, 1.37; 69–84 cases], while categories 2 and 4 have means near zero with unstable signs. A simple linear anxiety interpretation is not supported. Continuous warmth measures have training-specific quartile boundaries; their saved signed cells should be compared only when the intervals match.

What is fixed for the later election analysis

The later 2024 analysis preserves reported vote rather than nonvoter support, within-group weighting, the three nested information sets, training-only preprocessing, the frozen nine-setting grid, repeated holdout assessment, item-level SHAP, and signed response interpretation. Sample definitions, survey weights, wording, routing, and variable exclusions require fresh wave-specific audits. No cross-election harmonization, comparison, or final importance cutoff is established here.

Technical audit / reproducibility notes

Sources and sample construction

The source data are 39096-0001-Data.rda, object da39096.0001, with 17,545 rows and 1,490 variables. The local questionnaire and ICPSR codebook are 39096-Questionnaire.pdf and 39096-0001-Codebook-ICPSR.pdf. Exact paths, roles, MD5 and SHA-256 checksums are in source-manifest.csv. The earlier preparation and analysis files were inspected as context but are not sourced or used as a prepared dataset. The QMD contains all executable preparation, fitting, aggregation, and reporting functions.

Questionnaire PDF pp. 5 and 13 (printed pp. 3 and 11) establish registration and vote routing: S6=1, Q12=1 (“certain I voted”) or Q12=2 (“think I voted”), and Q14=1:5. Trump is Q14=1; Biden is 2, Jorgensen 3, Hawkins 4, someone else 5. Nonvoters receive support wording in the same source field and are excluded by the turnout condition. Voting mode (Q13) is asked only of certain voters; its structural missingness remains explicit. Latino membership uses MAIN_ETH_QUOTA=2, not an assumption that every Hispanic-ancestry response identifies the same sample.

BASELINE=1 restricts the run to the primary release. This resolves the weight scope choice explicitly before fitting. The raw field is labelled “Sept 1 2021 release”; it is the available primary-release indicator, not a claim that the public-use extract exactly reproduces the published primary sample.

stage	n
Raw records	17545
Latino quota, all releases	4301
Registered Latino, all releases	2368
Reported voters, all releases	2220
Valid choice, all releases	2220
Primary release, all races	14977
Primary Latino quota	3942
Registered primary Latino	2212
Reported primary Latino voters	2082
Valid choice: final sample	2082

The [official primary-sample methodology](#), linked from the [CMPS 2020 survey page](#), reports 14,988 primary interviews and 4,006 Latinos. The local primary indicator identifies 14,977 records, including 3,942 Latino-quota records (3,951 with RACE=2). These counts differ from the publication and from each other; no cases are added or reassigned to match published totals. The precise origin of the discrepancy remains unresolved. The primary methodology describes April 2–August 25, 2021 fieldwork, after the election and January 6 events.

The methodology’s pp. 1–2 specify within-racial-group raking to 2019 ACS one-year adult margins for age, gender, education, nativity, and ancestry, within plus or minus two percentage points, and recommend the weight for multi-variate analysis. The codebook labels `WEIGHT` “Primary sample weights ONLY”; `OS_WEIGHT` is an oversample weight. The latter is not substituted for this primary-only run. Weighting adults does not separately calibrate the selected registered voters.

Wave-specific exclusion and routing rules

The ledger covers every source variable before any fit, including masked, administrative, routing, and otherwise eligible fields. Common exclusions remove the outcome and selection fields, identifiers, dates, weights, release flags, uncoded/open text, masked fields, duplicate age categories, and routing-only metadata. No appended census-context fields are identified in this extract. Every remaining field is screened separately on the appropriate training data.

The following lists give eligible source fields removed at each step. Detailed reasons and codebook PDF pages remain in the ledger.

Outcome-proximate: Q2R1, Q2R2, Q2R3, Q2R4, Q2R5, Q2R6, Q2R7, Q2R8, Q2R9, Q2R10, Q15, Q47, Q48, Q49, Q50, Q51, Q52, Q53, Q54, Q55, Q56, Q96R10, Q96R11, Q96R12, Q98R8, Q98R9, Q130, Q195A1R1, Q195A1R2, Q195A1R3, Q195A1R4, Q195A2R1, Q195A2R2, Q195A2R3, Q195A2R4, Q195A2R5, Q195A2R6, Q195A2R7, Q195A2R8, Q195A2R9, Q195A2R10, Q195B1R1, Q195B1R2, Q195B1R3, Q195B1R4, Q195B2R1, Q195B2R2, Q195B2R3, Q195B2R4, Q195B2R5, Q195B2R6, Q195B2R7, Q195B2R8, Q195B2R9, Q195B2R10, Q196, Q197A1, Q197A2, Q197B1, Q197B2, Q198, Q199, Q200, Q201, Q272, Q323_BLW, Q323_API, Q323_NINH, Q324, Q325_BLW, Q325_API, Q325_NINH, Q326R1, Q326R2, Q326R3, Q326R4, Q326R5, Q326R6, Q326R7, Q340, Q341R1, Q341R2, Q504R11, Q504R12, Q509_Q511R3, Q575_Q581R1, Q575_Q581R2, Q575_Q581R3, Q575_Q581R4, Q575_Q581R5, Q575_Q581R6, Q575_Q581R7, Q614, Q615, Q684.

Direct partisan: Q1R17, Q1R18, Q17, Q18, Q19, Q21, Q22, Q23, Q24, Q25, Q26, Q27R1, Q27R2, Q27R3, Q27R4, Q27R5, Q27R6, Q27R7, Q27R8, Q27R9, Q27R10, Q27R11, Q27R12, Q27R13, Q27R14, Q27R15, Q27R16, Q27R17, Q27R18, Q27R19, Q28R1, Q28R2, Q28R3, Q28R4, Q28R5, Q28R6, Q28R7, Q28R8, Q28R9, Q28R10, Q28R11, Q28R12, Q28R13, Q28R14, Q28R15, Q28R16, Q28R17, Q28R18, Q28R19, Q225R10, Q225R11, Q287R6, Q294, Q301R1, Q301R2, Q301R3, Q301R4, Q301R5, Q301R6, Q301R7, Q315, Q316R7, Q334, Q560R4, Q743, Q744, Q745.

Election-fraud and January 6 items are withheld as explicitly election-framed responses, including news awareness (Q47) and the 2020 election tradeoff (Q684). General voting rules, political participation, nonspecific campaign contact, and electoral-college policy remain eligible. Candidate experiments include ranked/single-choice 2024 vignettes (Q195–Q201), state-assembly choice (Q272), candidate representation/trust (Q323–Q328), and the council conjoint (Q340–Q341). The candidate-feeling battery Q575_Q581 is withheld in full. Generic desired representative traits remain eligible except explicit shared-party preference. These are conceptual rules fixed before importance estimation.

The additional partisan exclusions include Q17–Q19 foreign-party attachment and its routed follow-ups; Q21–Q28 identity/history and party associations; party-antagonism priorities Q1R17/R18; party support/threat to one’s vision of society Q225R10/R11; party identity in Q287R6, Q294, Q301, Q315, Q560R4; shared-party candidate preference Q316R7;

partisan redistricting tradeoff Q334; and spouse’s identity/strength/leaning Q743–Q745. The entire 100-point Q301 allocation is withheld because its remaining components could reconstruct the party allocation. Ideology (Q43) remains available. The least-liked-group ranking includes Democratic Socialists and the Tea Party as named movements, not Democratic or Republican party identification. It and its piped follow-ups remain eligible, as do congressional-caucus evaluations and nonspecific campaign contact; these boundary decisions do not claim that the responses are unrelated to partisanship. No identifiable parental-party item was found. Racial, policy, religious, economic, and demographic opinions are not removed merely as party proxies.

Randomized target/wording fields are joined where needed, including immigrant health frames, representation vignettes, local-institution diversity, women’s stereotypes, government-level trust, and outsider/insider wording. Distinct source fields already tied to different wording remain distinct. The least-liked-group pipe is joined to Q607_Q610 responses. Rich conjoint profiles can produce rare joint categories; the unchanged rare-level rule may remove their encoded variation. Their policy eligibility does not guarantee retention. Placement-only metadata never enters as a separate predictor.

Measurement issues requiring review

ICPSR processing notes on codebook PDF p. 3 identify inconsistent wording for Q315 and Q603R2 between the questionnaire and data. Supplied response labels are retained; those codes receive review flags and no confident substantive name. Q315 is joined to its selected identity target and withheld from Model 3 because that target can reveal party identity. Q603R2 remains a component of the source-learning ranking question; its disputed target must not be interpreted as established. Truncated labels elsewhere remain linked to their source code and are flagged where full wording has not been verified. The response-level CSV preserves supplied category labels rather than guessing their completion. Source codes and questionnaire numbers differ in places: data Q614 is questionnaire item 613 on Trump immigration policies; data Q615 is item 614 on expressing that view. The verified crosswalk records these correspondences by wording, including the shifted discrimination battery.

Calendar-year factors Q742, Q806, and Q808 use their displayed years rather than stored level indices. Birth year, sliders and allocations remain numeric; Q504, Q603, and Q674 rating/rank categories remain categorical. Explicit special responses on other fields are not deleted. Duration fields retain their supplied categories, including “less than a year.” System missingness cannot distinguish every reason for nonresponse. Rare pooling may combine substantively meaningful categories; `rare-category-training-audit.csv` records the affected inner- and outer-training categories and counts.

Fitted-feature and reconstruction audit

All nine models passed source-variable exclusions, encoded-item membership, nesting within each seed, ideology retention, grid selection, and SHAP reconstruction. Maximum reconstruction error is **5.77316e-15**, below 10^{-6} . The weighted AUC implementation also passes an independent tied-score check.

Model	Seed	Option	Node	Inner_mtry	Final_mtry
1	202001	fraction20	20	1147	1174
2	202001	fraction20	5	1019	1038
3	202001	fraction20	5	974	992
1	202002	fraction20	20	1146	1169
2	202002	fraction20	5	1018	1034
3	202002	fraction40	5	1947	1976
1	202003	fraction40	5	2296	2348
2	202003	sqrt	5	71	72
3	202003	fraction20	5	975	992

The 40% upper boundary was selected in **2 of nine fits**. The grid was not expanded. Inner and final `mtry` counts differ when the training-specific encoded feature count differs.

Calibration diagnostics

Model	N	Prevalence	Prediction	Intercept	Slope
Model 1	521 [521, 521]	0.286 [0.277, 0.303]	0.284 [0.271, 0.309]	0.04 [-0.10, 0.12]	1.52 [1.32, 1.74]
Model 2	521 [521, 521]	0.286 [0.277, 0.303]	0.285 [0.280, 0.294]	0.02 [-0.04, 0.11]	1.76 [1.33, 2.33]
Model 3	521 [521, 521]	0.286 [0.277, 0.303]	0.297 [0.290, 0.307]	-0.09 [-0.13, -0.03]	1.58 [1.36, 1.74]

Values are mean [minimum, maximum]. An intercept near zero and slope near one indicate closer calibration. All nine slopes exceed one, suggesting that the fitted probability contrasts are too compressed relative to the observed holdout relationship. Model 3 also predicts a higher mean Trump share than observed in every split. The high AUC values therefore do not establish probability calibration. No calibration correction is fitted to these holdouts.

Reproduction and saved outputs

From this QMD's directory, run:

```
quarto render cmps2020_latino_trump_analysis.qmd --to pdf
```

R packages **ranger**, **treeshap**, **digest**, and **knitr**, Quarto, and XeLaTeX are required; package versions are recorded in **session-info.txt**. The source root defaults to `/Users/jessala/Desktop/Active/Github/ml-latino-vote-2020` and can be relocated with **CMPS2020_SOURCE**. The raw data and two local documentation checksums are asserted. An ordinary render recomputes all nine fits. Optional **CMPS2020_SCRATCH** holds fingerprinted aggregate-only checkpoints during layout verification; no external preparation or baseline result is required.

The **results/** directory contains aggregate sample, source, disposition, tuning, performance, calibration, importance, direction, feature, pooling, and validation records plus three top-50 figures. The QMD, PDF, results directory, and concise execution note are the persistent deliverables. Forest objects, respondent IDs, individual predictions and SHAP arrays are never exported. Scratch render files and aggregate checkpoints are removed after final verification.